

Aegivex AI - API Documentation

Property	Details
Version	1.0
Document Type	API Documentation
Project	OKX.AI Genesis Hackathon
Prepared For	Development Team
Prepared By	Aegivex AI Team

1. Overview

This document defines the REST API specification for Aegivex AI.

The backend is built using **FastAPI** and exposes secure REST APIs for authentication, AI interactions, blockchain security analysis, scan history, and user management.

Configuration	Value
Base URL	https://api.aegivex.ai/v1
Content Type	application/json
Authentication	JWT Bearer Token

2. Authentication APIs

2.1 Register User

Endpoint: POST /auth/register

Request:

```
{
  "name": "John Doe",
  "email": "john@example.com",
  "password": "StrongPassword123"
}
```

Response:

```
{
  "success": true,
  "message": "User registered successfully."
}
```

2.2 Login

Endpoint: POST /auth/login

Request:

```
{
  "email": "john@example.com",
  "password": "StrongPassword123"
}
```

Response:

```
{
  "access_token": "JWT_TOKEN",
  "token_type": "Bearer"
}
```

2.3 Logout

Endpoint: POST /auth/logout

Authorization Required: Bearer Token

Response:

```
{
  "success": true
}
```

2.4 Get Current User

Endpoint: GET /auth/me

Authorization Required: Bearer Token

Response:

```
{
  "id": "uuid",
  "name": "John Doe",
  "email": "john@example.com"
}
```

3. AI Chat API

Endpoint: POST /ai/chat

Authorization Required: Yes

Request:

```
{
  "prompt": "Is this wallet safe?"
}
```

Response:

```
{
  "response": "This wallet appears safe.",
  "risk_score": 12,
  "confidence": 96
}
```

4. Wallet Scanner API

Endpoint: POST /scan/wallet

Request:

```
{
  "wallet_address": "0x123..."
}
```

Response:

```
{
  "risk_score": 18,
  "risk_level": "Low",
  "summary": "Wallet activity appears normal.",
  "recommendation": "Safe to interact."
}
```

5. Token Scanner API

Endpoint: POST /scan/token

Request:

```
{
  "contract_address": "0xABC..."
}
```

Response:

```
{
  "token_name": "Example Token",
  "symbol": "EXT",
  "risk_score": 45,
  "risk_level": "Medium",
  "honeypot": false,
  "recommendation": "Proceed with caution."
}
```

6. Smart Contract Scanner API

Endpoint: POST /scan/contract

Request:

```
{
  "contract_address": "0xDEF..."
}
```

Response:

```
{
  "verified": true,
  "proxy_contract": false,
  "risk_score": 21,
  "risk_level": "Low",
  "recommendation": "No major risks detected."
}
```

7. Website Scanner API

Endpoint: POST /scan/website

Request:

```
{
  "url": "https://example.com"
}
```

Response:

```
{
  "trust_score": 92,
  "ssl_status": "Valid",
  "risk_level": "Low",
  "recommendation": "Website appears legitimate."
}
```

```
}
```

8. Transaction Explainer API

Endpoint: POST /scan/transaction

Request:

```
{  
  "transaction_hash": "0x987..."  
}
```

Response:

```
{  
  "summary": "This transaction requests unlimited token approval.",  
  "risk_score": 82,  
  "risk_level": "High",  
  "recommendation": "Avoid signing this transaction."  
}
```

9. Scan History API

Endpoint: GET /history

Response:

```
[  
  {  
    "scan_type": "wallet",  
    "risk_score": 15,  
    "created_at": "2026-07-22T10:30:00Z"  
  }  
]
```

10. Dashboard API

Endpoint: GET /dashboard

Response:

```
{
  "total_scans": 54,
  "wallet_scans": 20,
  "token_scans": 12,
  "contract_scans": 8,
  "website_scans": 9,
  "transaction_scans": 5
}
```

11. Notification API

Endpoint: GET /notifications

Response:

```
[
  {
    "title": "High Risk Transaction",
    "message": "A recent scan detected a high-risk approval request.",
    "is_read": false
  }
]
```

12. User Profile API

12.1 Get Profile

Endpoint: GET /profile

Response:

```
{
  "name": "John Doe",
  "email": "john@example.com",
  "joined": "2026-07-01"
}
```

12.2 Update Profile

Endpoint: PUT /profile

Request:

```
{
  "name": "John Doe",
  "profile_image": "avatar.png"
}
```

Response:

```
{
  "success": true
}
```

13. Error Responses

HTTP Status Code	Sample Response
400 Bad Request	{ "error": "Invalid input." }
401 Unauthorized	{ "error": "Authentication required." }
403 Forbidden	{ "error": "Access denied." }
404 Not Found	{ "error": "Resource not found." }

HTTP Status Code	Sample Response
429 Too Many Requests	{ "error": "Rate limit exceeded." }
500 Internal Server Error	{ "error": "Unexpected server error." }

14. Security Features

- **JWT Authentication:** Secure bearer tokens for session management.
- **HTTPS Only:** Enforced encrypted communication.
- **Rate Limiting:** Protection against DDoS and brute-force attacks.
- **Request Validation & Input Sanitization:** Prevention of injection vulnerabilities.
- **Secure Headers:** Enhanced HTTP response security.
- **API Key Protection:** Controlled service access.
- **CORS Configuration:** Configured cross-origin resource sharing rules.

15. API Versioning

Version	Status
v1	Current Version
v2 / v3	Future Planned Versions

Note: Older versions remain supported during migration periods.

16. Expected Outcome

The API provides a secure, scalable, and consistent interface between the frontend, AI engine, blockchain intelligence services, and database, enabling all Aegivex AI features through standardized REST endpoints suitable for production deployment.